

Advanced (Habanero)

- Q1.** Which of the following is a characteristic of a positive correlation?
(A) As x increases, y decreases
(B) As x increases, y increases
(C) As x decreases, y decreases
(D) As x increases, y remains constant
(E) No relationship between x and y
- Q2.** What is the purpose of a line of best fit?
(A) To find the correlation between two variables
(B) To predict future data trends
(C) To identify patterns in data
(D) To create a scatter plot
(E) All of the above
- Q3.** If a scatter plot shows a strong negative correlation, what can be said about the slope of the line of best fit?
(A) It will be positive
(B) It will be negative
(C) It will be zero
(D) It will be undefined
(E) It will be infinite
- Q4.** Which of the following is an example of a scatter plot with no correlation?
(A) A straight line with a positive slope
(B) A straight line with a negative slope
(C) A random collection of points
(D) A curve with a positive slope
(E) A curve with a negative slope
- Q5.** What is the equation of a line of best fit for the data points (1, 2), (2, 3), and (3, 4)?
(A) $y = x + 1$
(B) $y = x - 1$
(C) $y = 2x + 1$
(D) $y = 2x - 1$
(E) $y = x^2 + 1$
- Q6.** If a line of best fit has a slope of 2 and passes through the point (1, 3), what is its y -intercept?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q7.** What is the purpose of using a line of best fit to make predictions?
(A) To find the exact value of a data point
(B) To identify patterns in data
(C) To estimate the value of a data point based on trends
(D) To create a scatter plot
(E) To find the correlation between two variables
- Q8.** If a scatter plot shows a weak positive correlation, what can be said about the line of best fit?
(A) It will be a horizontal line
(B) It will be a vertical line
(C) It will have a small positive slope
(D) It will have a large positive slope
(E) It will have a negative slope

Open Ended Questions (FRQ)

- Q9.** Create a scatter plot for the data points (1, 2), (2, 4), (3, 6), and (4, 8). Then, draw a line of best fit and use it to predict the value of y when $x = 5$.
- Q10.** Explain the difference between a positive and negative correlation. Provide an example of each and describe how the line of best fit would be used to make predictions in each case.

Chef's Tips

- Tip 1: Emphasize the importance of labeling axes and title in scatter plots
- Tip 2: Encourage students to use technology to create and analyze scatter plots
- Tip 3: Remind students to always check their work and provide clear explanations for their answers